

LAB EXPERIMENT RECORD

SET 11: PRODUCT CATEGORY ANALYSIS

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To analyze product category sales data using a funnel chart, a sales table and a pie chart in R, and to build an interactive dashboard using Tableau.

Dataset: Product Categories

Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

PROGRAM 1: Funnel Chart of Sales by Product Category

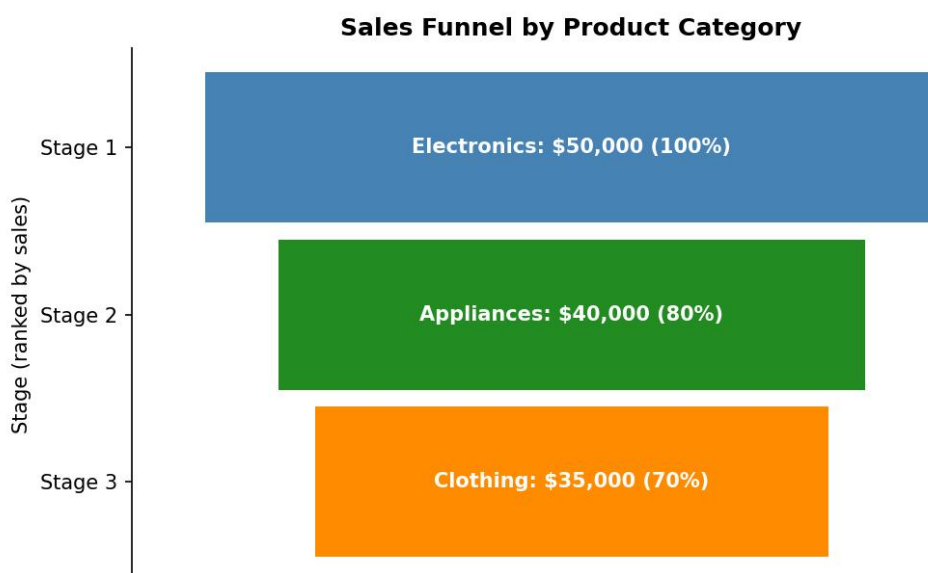
Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

```
library(ggplot2)

# Dataset
sales_data <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales     = c(50000, 35000, 40000)
)

# Funnel stages: categories ranked from highest to lowest sales
funnel <- sales_data[order(-sales_data$Sales), ]
funnel$Stage <- factor(paste("Stage", 1:3),
  levels = rev(paste("Stage", 1:3)))
funnel$Label <- paste0(funnel$Category, ": $",
  format(funnel$Sales, big.mark = ","),
  " (", round(funnel$Sales / max(funnel$Sales) * 100), "%)")

# Funnel chart: bars centred on the same axis
ggplot(funnel, aes(x = 0, y = Stage, width = Sales, fill = Category)) +
  geom_tile(height = 0.9) +
  geom_text(aes(label = Label), colour = "white", fontface = "bold") +
  scale_fill_manual(values = c(Electronics = "steelblue",
    Clothing = "darkorange",
    Appliances = "forestgreen")) +
  labs(title = "Sales Funnel by Product Category",
    x = NULL, y = "Stage (ranked by sales)") +
  theme_minimal() +
  theme(axis.text.x = element_blank(), panel.grid = element_blank(),
    legend.position = "none")
```



Output 1: Sales Funnel by Product Category (Funnel Chart)

Observation: The funnel ranks the categories from the highest to the lowest sales. Electronics is the first stage with \$50,000 (100%), Appliances is the second stage with \$40,000 (80% of Electronics) and Clothing is the last stage with \$35,000 (70% of Electronics). Sales narrow by \$10,000 (20%) from Electronics to

Appliances and by a further \$5,000 (12.5%) from Appliances to Clothing, so the total narrowing from the top stage to the bottom stage is \$15,000 (30%).

Note: The dataset contains only the total sales of each category and has no earlier conversion stages (such as visits or orders), so the stages of the funnel are the categories ranked by sales.

PROGRAM 2: Table of Sales for Each Product Category

Build a table to display the sales data for each product category. Label the table elements.

```
# Dataset with labelled columns
sales <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales     = c(50000, 35000, 40000)
)

# Add the share of each category in total sales (%)
sales$Percent <- round(sales$Sales / sum(sales$Sales) * 100, 1)

# Display the table
print(sales)

# Total sales of all categories
sum(sales$Sales)
```

Console Output

```
  Category Sales Percent
1 Electronics 50000      40
2  Clothing  35000      28
3 Appliances 40000      32

[1] 125000
```

Output 2a: Product Category Sales Table (R Console)

Category	Sales (\$)	Share of Total Sales
Electronics	50,000	40%
Clothing	35,000	28%
Appliances	40,000	32%
Total	125,000	100%

Output 2b: Sales Table with Totals

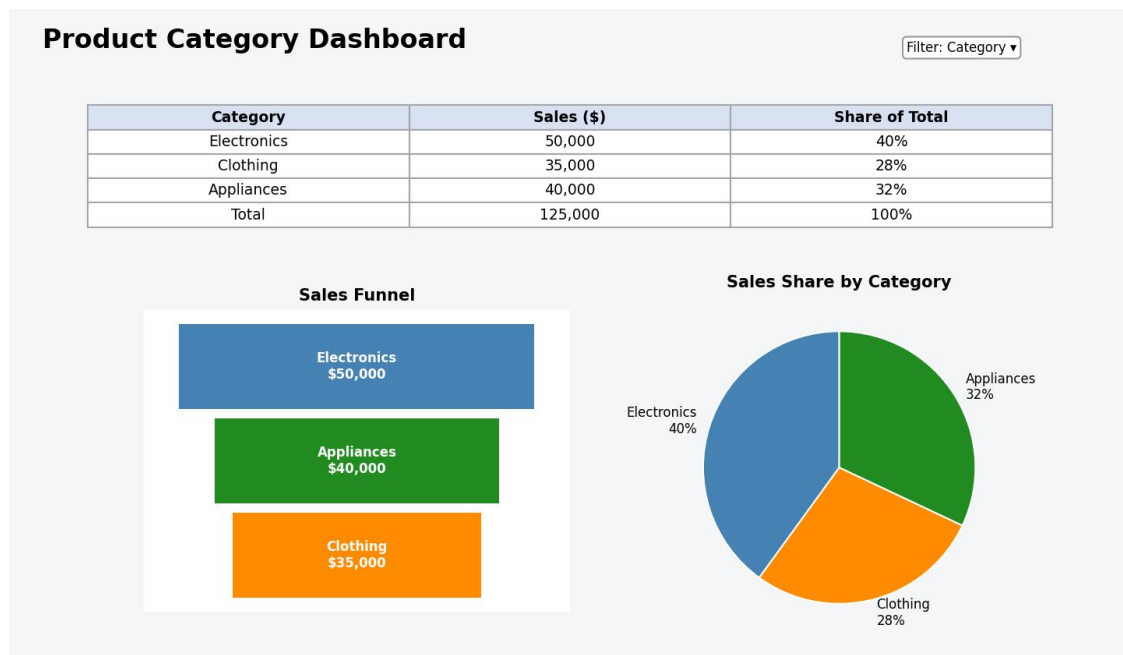
Observation: The table gives the exact sales of each category. Electronics has the highest sales (\$50,000, 40% of the total) and Clothing the lowest (\$35,000, 28%). Total sales are \$125,000, an average of about \$41,667 per category, and Electronics sells about 43% more than Clothing.

PROGRAM 3: Interactive Product Category Dashboard in Tableau

Develop a Tableau dashboard combining the pie chart, funnel chart and the table for interactive exploration of product category data.

Steps in Tableau

1. Save the data (Category, Sales) as an Excel or CSV file and connect to it in Tableau (Connect > Microsoft Excel / Text file).
2. Sheet 1 (Funnel Chart): Drag Category to Rows. Create a calculated field named "Funnel Position" with the formula $-\text{SUM}([\text{Sales}])/2$ and drag it to Columns. Change the Marks type to Gantt Bar and drag SUM(Sales) to Size, so each bar is centred. Sort Category in descending order of SUM(Sales), drag Category to Color and SUM(Sales) to Label, and hide the x-axis header. Rename the sheet "Sales Funnel".
3. Sheet 2 (Pie Chart): Change the Marks type to Pie. Drag Category to Color, SUM(Sales) to Angle, and add Category and SUM(Sales) to Label, using a Percent of Total quick table calculation for the percentage. Rename the sheet "Sales Share".
4. Sheet 3 (Table): Drag Category to Rows and SUM(Sales) to Text. Drag SUM(Sales) to Text again and apply Percent of Total. Use Analysis > Totals > Show Column Grand Totals. Rename the sheet "Sales Table".
5. Create a new Dashboard (Dashboard > New Dashboard), set the size to Automatic, and drag the three sheets onto the dashboard area.
6. Add interactivity: select the pie chart and the funnel chart and click Use as Filter, so clicking a slice or a stage filters the other views. Add a Category filter (Show Filter > Apply to Worksheets > All Using This Data Source).
7. Add the title "Product Category Dashboard" and test the filters in Presentation Mode.



Output 3: Interactive Product Category Dashboard (Tableau)

Observation: The dashboard shows the sales table, the funnel and the pie chart together. Clicking a pie slice (for example, Electronics) or a funnel stage highlights or filters the same category in the other views, and the Category filter lets users compare any selection of categories.

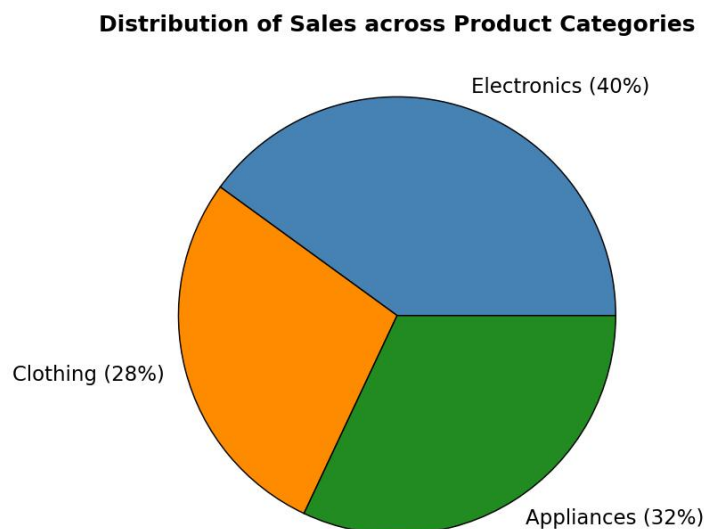
PROGRAM 4: Pie Chart of Sales across Product Categories

Create a pie chart to represent the distribution of sales across product categories. Include labels.

```
# Dataset
category <- c("Electronics", "Clothing", "Appliances")
sales     <- c(50000, 35000, 40000)

# Percentage of total sales for each category
percent <- round(sales / sum(sales) * 100, 1)
labels  <- paste0(category, " (", percent, "%)")

# Pie Chart
pie(sales,
    labels = labels,
    col    = c("steelblue", "darkorange", "forestgreen"),
    main   = "Distribution of Sales across Product Categories")
```



Output 4: Distribution of Sales across Product Categories (Pie Chart)

Observation: Electronics contributes the largest share of sales (40%), followed by Appliances (32%) and Clothing (28%). Electronics and Appliances together account for 72% of the total sales of \$125,000.

Procedure

1. Define the category and sales values in R as vectors and a data frame.
2. Rank the categories by sales with `order()` and use `ggplot2` with `geom_tile()` and `geom_text()` to draw the labelled funnel chart.
3. Add a share column with `round()` and `sum()`, and display the labelled table with `print()`.
4. Load the sales file into Tableau and build the funnel chart (Gantt Bar), pie chart and table worksheets.
5. Combine the worksheets in a Tableau dashboard and add filters and actions for interactivity.
6. Calculate the percentage of each category and use `pie()` with labels to draw the pie chart in R.

RESULT:

The product category sales data was successfully analyzed using a funnel chart, a sales table and a pie chart in R. An interactive Tableau dashboard was also created, combining the funnel chart, pie chart and table so that users can explore the sales of each product category.

Key Functions and Tools Used

Function / Tool	Purpose
<code>data.frame()</code> / <code>order()</code>	Creates the sales table and ranks the categories.
<code>ggplot()</code> / <code>geom_tile()</code> / <code>geom_text()</code>	Creates the funnel chart with labelled stages.
<code>sum()</code> / <code>round()</code>	Calculates total sales and the percentage share.
<code>print()</code>	Displays the sales table.
<code>pie()</code> / <code>paste0()</code>	Creates the pie chart with percentage labels.
Tableau Gantt Bar	Builds the funnel chart from centred bars.
Tableau Dashboard	Combines worksheets with filters and actions.